



Anticipatory action for El Niño in Latin America and the Caribbean

Strengthening local capacities and partnerships
to protect livelihoods

Summary

This good practice factsheet documents the Food and Agriculture Organization of the United Nations' (FAO's) anticipatory action (AA) in response to El Niño across eight countries in Latin America and the Caribbean between June 2023 and February 2024. Drawing on seasonal and climate forecasts, agricultural calendars, territorial risk analyses and predefined crisis timelines, FAO implemented AAs to protect food security and agricultural livelihoods. The actions targeted 31 873 households engaged primarily in subsistence agriculture and small-scale livestock production, with a strong focus on Indigenous Peoples and female-headed households.

The good practice factsheet highlights how AAs – such as the provision of agricultural and livestock inputs, cash transfers, water harvesting and irrigation systems, climate-resilient seeds, animal health brigades and family productive spaces – contributed to measurable improvements in food security, livelihoods and resilience. Implementation relied on close collaboration with governments and local partners, with these synergies proving to be essential to the success and sustainability of the initiative. Results include significant yield increases, reductions in production losses and animal mortality, and improved food security indicators among beneficiary households. Return on investment (ROI) analysis shows that, across the region, AA generated an average return of USD 2.34 for every dollar invested, demonstrating both effectiveness and value for money. Beyond immediate impacts, the practice strengthened local and national capacities and supported the institutionalization of AA within agroclimatic risk management and public policy frameworks, reinforcing sustainability and the potential for replication and scale-up across diverse contexts.

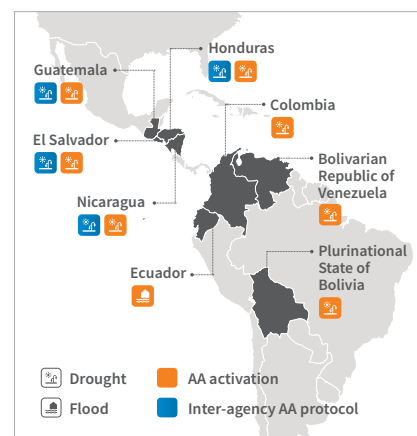
The experience confirms that AA is most effective when it is anchored in robust early warning systems, flexible financing and strong operational preparedness at country and regional levels. Evidence-based targeting, combined with context-adapted technologies and close coordination among local actors and governments, is critical to reaching the most vulnerable households in a timely and efficient manner. By developing local capacities and fostering collaboration between communities and institutions, this good practice shows how anticipatory approaches can generate sustainable resilience outcomes and be scaled across diverse agroecological and socioeconomic contexts.

Key facts



Geographical coverage

Approximately 300 communities, municipalities and cantons in eight countries: Plurinational State of Bolivia, Colombia, Ecuador, El Salvador, Guatemala, Honduras, Nicaragua and the Bolivarian Republic of Venezuela



Note: Please refer to the disclaimer on the last page for the names and boundaries used in this map.

Source: United Nations Geospatial. 2025. Map of the World. In: *United Nations*. [Cited 17 April 2026]. <https://www.un.org/geospatial/content/map-world-1>



Period of implementation

June 2023–February 2024



Target group

More than 100 000 people (31 873 households) engaged in subsistence agriculture and supplemented their income with poultry or livestock, with a focus on the most vulnerable groups, including Indigenous Peoples and female-headed households



© FAO/Alcides Ramirez

Context

In 2024, extreme climate events and increasing climate variability were the primary drivers of food insecurity and malnutrition in Latin America and the Caribbean. The El Niño–Southern Oscillation (ENSO) – whose warm phase is known as El Niño – is associated with the abnormal warming of Pacific waters. This phenomenon significantly impacts wind patterns, sea surface temperatures and precipitation in the tropical Pacific (UNDRR, 2024).

In November 2023, the National Oceanic and Atmospheric Administration, the International Research Institute for Climate and Society, and the World Meteorological Organization (WMO) officially declared the onset of El Niño, which peaked between December 2023 and April 2024, concluding in June 2024. According to the WMO, “between January and September 2024, the average global temperature exceeded the pre-industrial baseline by 1.54 (±0.13) °C (WMO, 2025).

Across the region, El Niño triggered severe drought and above-average rainfall, disrupting agricultural cycles and food systems. These climate shocks were compounded by socioeconomic inequalities, informal labour and limited social protection coverage, further increasing the vulnerability of rural populations that rely predominantly on farming for their livelihoods.

In the dry corridor of Central America, irregular rainfall caused significant impacts. In Guatemala, bean production fell to below-average levels, while maize and bean prices rose by 19 percent and 61 percent, respectively, due to

insufficient water for irrigation (FAO, 2023). In Nicaragua, an initial drought followed by late-season rains reduced maize yields by at least 25 percent, affecting both subsistence households and commercial agricultural production (FAO, 2023). In Honduras, the drought delayed the first season planting by a month, increasing agricultural costs and reducing yields (FEWS NET, 2026).

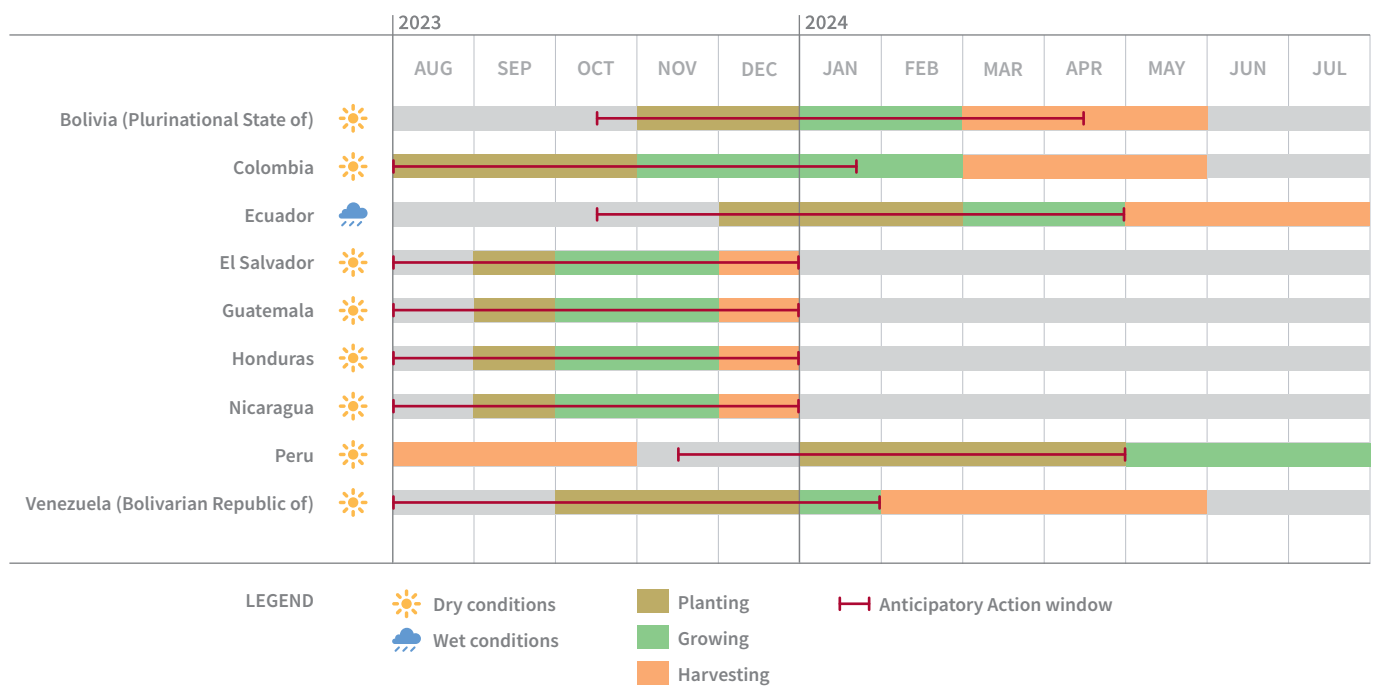
Within this context, FAO and partners implemented coordinated AAs in Latin America and the Caribbean – resulting in significant improvements in food security, livelihood protection and resilience among rural communities affected by El Niño-related extreme climate events. For FAO, the AA approach involves linking pre-emptive measures to mitigate the impacts of adverse events before they fully materialize, thereby protecting lives and livelihoods and reducing humanitarian needs. Its effectiveness depends on predefined thresholds (triggers) that enable rapid disbursement of pre-allocated funds. Activities are activated through **flexible financing mechanisms such as the Special Fund for Emergency and Resilience Activities (SFERA)**. Drawing on meteorological forecasts and impact analyses of El Niño on food security and agriculture – supported by various partners including the European Union, the governments of Belgium and Norway, and the United Nations Central Emergency Response Fund, through SFERA – FAO facilitated AAs that helped safeguard food security and agricultural livelihoods of 31 873 households across the region.

Methodological approach

AAs in the region were activated based on several key elements:

- ENSO forecasts, confirming the presence of the climatic phenomenon;
- anomalous rainfall patterns, including predictions of below-average precipitation in drought-prone areas and above-average precipitation in flood-prone zones (Coastal Niño);
- seasonal calendars for agricultural and livestock production (Figure 1);
- territorial risk analysis and identification of vulnerable populations; and
- crisis timelines.

Figure 1. Seasonal agricultural calendars for priority countries in Latin America and the Caribbean



Source: FAO. 2023. *El Niño: Anticipatory Action and Response Plan (August–December 2023). Mitigating the expected impacts of El Niño-induced climate extremes on agriculture and food security*. Rome. <https://doi.org/10.4060/cc7267en>

Implemented AAs included:

- provision of agricultural and livestock inputs and distribution of drought- and flood-resistant seeds;
- unconditional cash transfers;
- rehabilitation of water sources and installation of irrigation systems;
- deployment of animal health and nutrition brigades,
- establishment of family productive spaces (such as local family vegetable gardens); and
- the creation of community seed banks and contingency mutual funds.

The interventions were designed with a culturally adapted approach, incorporating alert dissemination and technical capacity development (soil regeneration and efficient water use) while involving Indigenous Peoples and local actors. This ensured relevance, sustainability, and community ownership through validation and consent processes.



In terms of targeting and selection criteria, priority was given to the most vulnerable households, including those who:

- rely mainly on agriculture for their livelihoods;
- are highly affected by food insecurity;
- cultivate no more than three hectares of land and/or own no more than five livestock;
- have limited access to water for irrigation or livestock;
- have dependents and/or are headed by women; and
- are known for their commitment to the community.

AAs in the region was implemented in collaboration with local partners and governments, each fulfilling a distinct role. Local partners – including community organizations – contributed their knowledge of the area and facilitated active community participation, ensuring that initiatives were responsive to local needs and priorities. Government counterparts provided the necessary regulatory framework and funding, and ensured that public policies supported the sustainable development of the projects. The synergy between local partners and governments was essential for the success and sustainability of the initiative. In areas with Indigenous Peoples, obtaining free, prior and informed consent was essential for adapting the implementation strategy while improving coordination and monitoring at local level.

Return on investment

Context analyses were tailored to identify the specific social, political and economic dynamics of each territory. The regional analysis integrated context-specific factors, types of impact and resource availability for the development of impact assessment tools and methodologies. For consistency, these tools were aligned with monitoring, evaluation, accountability and learning (MEAL) frameworks, and the procedures of the FAO Office of Emergencies and Resilience. Primary data collection methodologies were harmonized across countries to ensure consistency between baseline and endline measurements. Digital data collection tools – such as Kobo Collect – were used to support standardized data gathering and were adapted to each local context. This allowed for a systematic analysis of the relevance and effectiveness of interventions in meeting the indicators set for each intervention. In Colombia, El Salvador, Guatemala, Honduras and Nicaragua, impact analyses primarily applied quantitative methodologies, including monetary investment metrics, such as ROI, to measure the impact of interventions. In contrast, qualitative analysis tools, such as testimonial-based evidence collection, were prioritized in the Plurinational State of Bolivia and the Bolivarian Republic of Venezuela.



The ROI tool compares the direct benefits of the project against its costs. Furthermore, it assesses the intervention's impact on various food security indicators by considering the following:

- **comparison of inputs, production systems and yields** between participating and non-participating households to measure the net monetary benefits of the project;
- **comparative analysis of food security indicators** to evaluate differences in conditions between beneficiary and non-beneficiary households;
- **insights from focus group interviews** to understand perceptions regarding the relevance, timing, benefits and any unintended negative effects of the interventions; and
- **field monitoring** to validate the impact of activities on the ground.

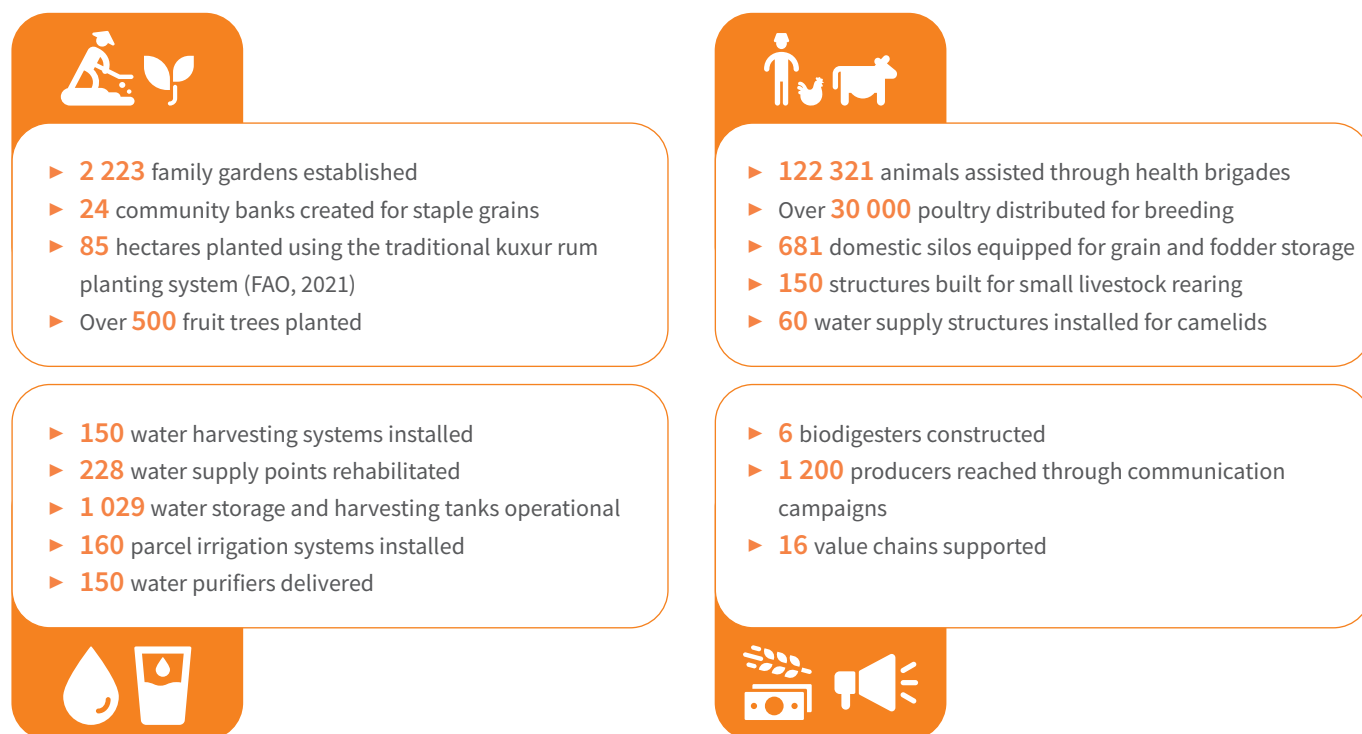
Throughout the data collection process, key data protection principles were observed. These included collecting the minimum data necessary, obtaining informed consent (with attention to community-led protection initiatives in Indigenous territories), ensuring transparency regarding the use of information and anonymizing records through unique household identification codes. Internal data management adhered to the 'do no harm' principle and complied with established protocols for secure data storage and confidentiality.

The registration and identification of participants were central to ensuring the quality of information regarding the delivery of supplies, records and assistance received. This process required continuous effort and analysis throughout data collection.



Results and impacts

While results varied depending on community needs, prioritized threats (droughts and floods), and the specific social, economic and environmental contexts, the regional analysis highlights the following achievements:



Beyond immediate results, these actions contributed to the institutionalization of the AA approach by strengthening local and national capacities. This has supported the integration of AA into public policies and agroclimatic risk management frameworks, enhancing the resilience and sustainability of agrifood systems across the region.

In addition, implementing AAs across Latin America and the Caribbean yielded significant improvements in food security, livelihood protection and the resilience of the most vulnerable rural communities. These efforts proved effective in mitigating the adverse effects of extreme climate events associated with El Niño, demonstrating the value of proactive and context-specific interventions.

Beneficiary households showed substantial improvements in food security, including increases of up to 10 points in the food consumption score and reductions of up to 6 points in the reduced Coping Strategies Index compared to control groups. Additionally, analysis using the Food Insecurity Experience Scale – over a 30-day period – revealed a lower severity of food insecurity in beneficiary households following the climatic event.

At regional level, over 100 000 people from at least 300 rural communities directly benefited from AAs. In the productive sector, yield increases ranging from 25 to 300 percent were recorded for key crops such as maize, beans and sorghum, alongside reductions in production losses of up to 84 percent. In the livestock sector, technical assistance helped reduce animal mortality by



up to 4 percentage points, improve livestock body conditions and increase milk production in targeted areas.

The average ROI across the region was estimated at USD 2.34 for every dollar invested in AA, with country- and intervention-specific variations ranging from USD 1.36 to USD 2.98, illustrating the intervention's ability to deliver tangible and sustainable results. These ROI results not only demonstrate the effectiveness of the investments but support its continuity and long-term impact.

As demonstrated in previous experiences related to AAs (FAO, 2018; FAO, 2019a; FAO, 2019b; FAO, 2024; WFP, 2020), acting before a crisis unfolds helps reduce humanitarian needs, preventing the escalation of extreme climate impacts in both urgency and magnitude. This approach also minimizes the demand for emergency assistance, reinforcing the protection of the well-being and dignity of vulnerable communities.

By involving local actors in both planning and implementation, a greater sense of ownership and commitment was fostered, enhancing the likelihood that benefits would be sustained over time.



Sustainability

The sustainability of AAs implemented across the region is reflected in the tailoring of the standardized system to context specificities, and the capacity of AAs to generate long-term benefits.

The coordinated collaboration and inclusive approach strengthened the relevance of the interventions, ensuring that strategies were aligned with the specific needs and contexts of each community. By involving local actors in both planning and implementation, a greater sense of ownership and commitment was fostered, enhancing the likelihood that benefits would be sustained over time.

Sustainability is measured not only by short-term economic gains but also by the ability of communities to maintain and expand these achievements over time, reflecting a deeper process of social and economic transformation. The diversity of results demonstrates that these interventions not only addressed immediate community needs but also contributed to long-term resilience in the face of climate and economic challenges.

Replicability and upscaling

Various local practices – such as the recovery of drought- and flood-tolerant seeds, planting schedules aligned with lunar cycles and ecological bioindicators, grain storage techniques, behavioural management of crops and the preparation of alternative and supplementary animal feed – have been applied across diverse social, economic and environmental contexts in the region. Building on this vast experience, the process of AA can be scaled up.

Additionally, the process of knowledge transfer and capacity development in AA has been facilitated through methodologies and tools, including the establishment of agroclimatic tables, dissemination of weather bulletins and the organization of workshops. These efforts have significantly enhanced the knowledge and preparedness of participating families and institutional actors. The following are key factors for replicability and upscaling:

- **Local knowledge:** Conducting thorough assessments of the local context – including cultural, social, economic and environmental factors – is essential. Understanding community needs, capacities and existing practices

ensures that adaptation efforts are relevant and effective. Involving local stakeholders, including communities, authorities and organizations, fosters ownership and increases the likelihood of successful implementation.

- **Flexibility and adaptability:** Developing a flexible framework that can be tailored to different contexts is crucial. This includes openness to modifying methodologies, tools and approaches based on local circumstances. Ensuring that practices are resilient to varying environmental conditions and socioeconomic situations enables successful adaptation across diverse settings.
- **Capacity development:** Providing training and resources for local implementers and communities enhances their capacity to adopt and adapt AA practices effectively. This may include workshops, manuals and ongoing technical support. Creating platforms for sharing knowledge and experiences fosters learning and innovation as different regions implement the practice.





Key learning

1. **AA can be institutionalized through public policies and risk management frameworks.** FAO's interventions in Colombia and the Bolivarian Republic of Venezuela, carried out in collaboration with the Coordination Centre for the Prevention of Natural Disasters in Central America, demonstrate that embedding AA into sectoral plans and regulations enhances sustainability and scalability. Integrating AA into agriculture, environment and civil protection policies – alongside predictable funding, clear activation protocols and early warning systems – ensures timely and effective responses.
2. **Operational preparedness and flexible funding mechanisms must be prioritized to enable rapid response.** Having AA plans, pre-agreements with partners and ready-to-activate logistics in place at country level is essential. Access to flexible funding – linked to early warning systems – enables rapid response once thresholds are reached, bridging the gap between preparedness and action. Mechanisms such as rapid response funds, humanitarian prefinancing or integration with shock-responsive social protection are key enablers.
3. **Evidence-based targeting and logistics planning must be strengthened to reach the most vulnerable.** Community diagnostics, harmonized baseline data and vulnerability analyses enabled FAO to prioritize support for the most at-risk households. Challenges in accessing remote areas highlighted the importance of preplanned logistical routes to ensure timely delivery of assistance.
4. **Operational flexibility and context-adapted technologies should be promoted to enhance impact.** Deploying resilient seeds, water harvesting systems, bio-inputs and traditional techniques led to improved yields and sustainability. Supporting local innovation and traditional knowledge not only enhances impact but also fosters community ownership and increases the potential for scaling.
5. **Multilevel governance and intersectoral coordination must be reinforced to ensure effective implementation.** The experience showed that active involvement of national, subnational and Indigenous authorities is essential to ensuring territorial coordination, community acceptance and the timely delivery of supplies. Establishing governance structures in advance – with defined roles, technical committees and shared protocols – strengthens preparedness and response capacities.
6. **MEAL mechanisms should be consolidated to ensure accountability and continuous learning.** Implementing MEAL strategies using tools and platforms such as Kobo Collect, while upholding data protection principles and community-led protection initiatives, improves transparency and quality. Capturing good practices and lessons learned through established knowledge management processes supports ongoing learning and accountability.



© FAO/Alcides Rodríguez

Testimonies

Sergio Cristino Castillo, a farmer from El Rebalse Community in San Antonio de Flores (Choluteca Department, Honduras) recalls how he was initially skeptical about the forecast: *“People were saying a drought was coming, but I didn’t believe it.”* His perspective changed after visits from FAO and the International Federation of Red Cross and Red Crescent Societies, which helped him understand the importance of preparing for the impact of the climate phenomenon. With the introduction of new agricultural techniques, Sergio and his family began to see tangible improvements in their harvest: *“Before, we planted without following any technique. Now, with the furrows and the support we received, the plants are growing well. We are going to have corn at home, which we didn’t have before,”* he explained with satisfaction. Beyond immediate gains, Sergio emphasized the importance of sustainability and self-reliance. His family is saving for the next harvest with the goal of continuing to produce independently.

In the Plurinational State of Bolivia, AAs proved to be pivotal to the livelihoods of farmers from the El Choro Municipality in Oruro Department.

For over five years, Ayllu Sora Chico, a livestock owner, was unable to resolve the issue of water scarcity in his area. As part of FAO’s AA activation to mitigate the impacts of El Niño-related drought, the farming family received a portable water reservoir, enabling them to provide water for their animals, when and where it is needed most.

Among the AAs activated in the country, Rosmery Anave most appreciated the application of bioinputs. On the use of Tricobal – a bioinput for seed treatment, designed to control soil-borne pathogens – in her plot, she says: *“For me, it has worked. The quinoa we planted with the Tricobal has already sprouted. The others we planted without Tricobal have not sprouted yet.”*

Another farmer, Edwin Churata, emphasized the impact of AAs on animal health, explaining the importance of reinforcing livestock diets. This was the family’s first experience preparing feed, thus ensuring its availability for their animals and safeguarding their original pastures: *“We introduced mineral salts, flour and molasses so that the dry grass had better properties, allowing it to survive better and feed the livestock more effectively.”*



© FAO/Alcides Rodríguez



© FAO/Raúl Pérez

References

- FAO.** 2018. *Horn of Africa: Impact of Early Warning Early Action. Protecting pastoralist livelihoods ahead of drought.* Rome. <https://openknowledge.fao.org/handle/20.500.14283/ca0227en>
- FAO.** 2019a. *Madagascar: Impact of Early Warning Early Action. Protecting farming livelihoods from drought and food insecurity.* Rome. <https://openknowledge.fao.org/handle/20.500.14283/ca3933en>
- FAO.** 2019b. *The Sudan: Impact of Early Warning Early Action. Protecting agropastoralist livelihoods ahead of drought.* Rome. <https://openknowledge.fao.org/handle/20.500.14283/ca4653en>
- FAO.** 2020. Sistema agroforestal Kuxur Rum, buenas prácticas agro para resiliencia, Corredor Seco Centroamerica [video]. [Cited 4 May 2026]. <https://www.youtube.com/watch?v=KliZQuYmbWU>
- FAO.** 2023. *The Impact of Disasters on Agriculture and Food Security 2023 – Avoiding and reducing losses through investment in resilience.* Rome. <https://doi.org/10.4060/cc7900en>
- FAO.** 2024. *Anticipatory action: Annual report 2023.* Rome. <https://openknowledge.fao.org/handle/20.500.14283/cd2966en>
- FEWS NET (Famine Early Warning Systems Network).** 2026. Honduras: Latest food security analysis. In: *FEWS NET*. Washington, DC. [Cited 4 May 2026]. <https://fews.net/latin-america-and-caribbean/honduras>
- UNDRR (United Nations Office for Disaster Risk Reduction).** 2024. Desarrollo de La Nina en el 2024: condiciones actuales, impactos esperados y acciones preparatorias. In: *UNDRR*. Geneva, Switzerland. [Cited 4 May 2026]. <https://www.undrr.org/es/event/desarrollo-de-la-nina-en-el-2024-condiciones-actuales-impactos-esperados-y-acciones>
- WFP (World Food Programme).** 2020. *The evidence base on Anticipatory Action.* Rome. <https://www.wfp.org/publications/evidence-base-anticipatory-action>
- WMO (World Meteorological Organization).** 2025. State of the Global Climate 2024. Geneva, Switzerland. <https://library.wmo.int/idurl/4/69455>

Acknowledgements

This publication was developed by the Emergencies and Resilience team in the FAO Regional Office for Latin America and the Caribbean, and written by Angela Blanco, Alexis Vallejo, Xabier Garay, Mariela Segura and Sebastian Cortinez, in collaboration with the Anticipatory Action teams in the Plurinational State of Bolivia, Colombia, Ecuador, El Salvador, Guatemala, Honduras and the Bolivarian Republic of Venezuela, and Lucia Palombi from the Knowledge Platform on Emergencies and Resilience, FAO Office of Emergencies and Resilience. The factsheet was edited by Paige McGreevy and designed and laid out by Anneta Bou Saleh.

Partners

Resource partners

- United Nations Central Emergency Response Fund through SFERA
- German Federal Foreign Office
- Swedish International Development Cooperation Agency
- Belgian Agency for International Cooperation
- Government of Canada

Technical partners

- **Plurinational State of Bolivia:** Ministry of Rural Development and Lands, Ministry of Environment and Water and local governments
- **Central American region:** German Red Cross
- **Colombia:** Ministry of Agriculture and Rural Development, the National Unit for Disaster Risk Management and local governments
- **Ecuador:** Ministry of Agriculture and Livestock and local governments
- **El Salvador:** General Directorate of the Observatory of Threats and Natural Resources and local governments
- **Guatemala:** Oriente University Centre and local governments
- **Nicaragua:** Nicaraguan Institute of Agricultural Technology and local governments
- **Honduras:** Pespirens Development Association, Environmental Municipal Unit, Municipal Unit of Integral Risk Management and local governments
- **Bolivarian Republic of Venezuela:** Promotion of Popular Development and local governments

Contact

FAO Regional Office for Latin America and the Caribbean
FAO-RLC@fao.org
fao.org/americas/en
Santiago, Chile

Office of Emergencies and Resilience
Anticipatory-Action@fao.org
KORE@fao.org
fao.org/in-action/kore
Rome, Italy

Food and Agriculture Organization
of the United Nations

This product, developed with the support and technical guidance of the Knowledge Platform on Emergencies and Resilience (KORE) in FAO's Office of Emergencies and Resilience, is available on its **portal**. KORE provides normative and methodological guidance and supports the generation of learning, documentation of good practices and dissemination of evidence-based knowledge to inform strategic and programmatic decision-making.

Required citation:

FAO. 2026. *Anticipatory action for El Niño in Latin America and the Caribbean – Strengthening local capacities and partnerships to protect livelihoods*. Rome.
<https://openknowledge.fao.org/handle/20.500.14283/CD9804EN>

The designations employed and the presentation of material in this information product do not imply the expression of any opinion whatsoever on the part of the Food and Agriculture Organization of the United Nations (FAO) concerning the legal or development status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries.



Some rights reserved. This work is made available under the Creative Commons Attribution - 4.0 International licence (CC BY 4.0)